

PATCHCORD RUN KEY PERSISTENCE HUNT

Windows registry hunt for the source-observed BeaconBrowserHijack current-user startup value.

HUNTING QUERY

CONFIDENCE: MEDIUM

SOURCE-CONFIRMED BEHAVIOR

retain facts, isolate interpretation

DeviceRegistryEvents

direct telemetry boundary

HUNT · VALIDATE · TUNE

no automatic malicious verdict

REPORT DATE

17 August 2026

VERSION

v1.0

PRIMARY TELEMETRY

Microsoft Defender XDR ·
DeviceRegistryEvents

ATT&CK

T1060

Evidence-led dossier. Query status remains an untested
implementation sketch until executed in the target environment.

INDEPENDENT RESEARCH FORMAT · CONTROLLED UI/UX MASTER



EXECUTIVE ASSESSMENT

Decision-grade summary for detection engineering and SOC operations.

ASSESSMENT

EXECUTIVE SUMMARY

Hunt the exact source-observed Run-key value.

Acronis TRU reports that PATCHCORD writes its executable path to BeaconBrowserHijack under the current user Run key. DeviceRegistryEvents directly records the key, value name, data and initiating process.

OPERATIONAL VALUE

Selective persistence evidence with a clear attribution boundary.

The exact value name sharply limits benign overlap and produces an actionable file/process pivot. It remains a mutable campaign artifact and must not be treated as proof of PATCHCORD or APT36.

8.9

01

OPERATIONAL VALUE



Medium

02

DETECTION CONFIDENCE



Hunting

03

OPERATIONAL READINESS



SOURCE-CONFIRMED FACTS

Observed behavior

- Published 13 August 2026; suspicious installer discovered in June 2026.
- PATCHCORD writes BeaconBrowserHijack under the current-user Run key.
- The stored data is the implant executable path; shortcut hijacking follows.

ANALYTICAL ASSESSMENT

What the evidence supports

- The exact Run-key value is a selective persistence pivot.
- RegistryValueData supplies the file path for follow-up.
- Artifact mutability prevents a durable Production classification.

DETECTION ASSUMPTIONS

What must be validated

- The Windows endpoint is onboarded to Defender for Endpoint.
- DeviceRegistryEvents captures the write and its initiating process.
- Approved software does not use the exact value name.

Confidence boundary

Acronis provides reverse-engineered malware behavior, not tenant Defender events. A query match does not prove the file identity, shortcut hijacking, execution, C2 activity, compromise scope or APT36 attribution.



THREAT MECHANICS

Source-observed chain and explicit observability boundaries.

ATTACK FLOW

01

Deliver installer

A themed Inno Setup package contains TMS_Launcher.exe.

02

Run implant

PATCHCORD initializes and hides its console.

03

Write Run value

BeaconBrowserHijack stores the implant path.

04

Hijack shortcuts

Browser shortcuts are backed up and rewritten.

05

Launch browser

The implant starts first and opens the real browser.

Observable core

A registry creation or modification event for BeaconBrowserHijack under the current-user Windows Run key, with value data and initiating-process context.

Non-observable boundary

Shortcut target content, preserved icon, successful startup execution, C2 tasking, in-memory shellcode and actor attribution.

DETECTION HYPOTHESIS

Campaign persistence should leave an exact registry artifact.

If PATCHCORD installs the documented persistence, DeviceRegistryEvents should record the BeaconBrowserHijack value under the current-user Run key. Legitimate activity can be separated using exact value name, stored path ownership, signer, hash, initiating process and shortcut evidence.

REQUIRED TELEMETRY

DeviceRegistryEvents

- Microsoft Defender for Endpoint registry telemetry.
- DeviceRegistryEvents.
- RegistryKey and RegistryValueName.

OPTIONAL ENRICHMENT

Context that improves disposition

- RegistryValueData file path and previous value.
- Initiating process, account, SHA1 and unique ID.
- File origin, signer, shortcut and network evidence.

UNAVAILABLE HERE

Do not infer

- Shortcut content and target arguments.
- Proof of startup execution or C2 success.
- Proof of PATCHCORD or APT36 attribution.



TELEMETRY & EVIDENCE MODEL

Minimum data contract and quality gates for a defensible hunt.

DATA CONTRACT

PRIMARY DATA SOURCE

Microsoft Defender XDR · DeviceRegistryEvents

DeviceRegistryEvents documents creation and modification of registry entries plus key, value name/data, device identity and initiating-process fields.

EVIDENCE BOUNDARY

A matching event is a lead, not a verdict.

ActionType is projected but not filtered because supported values must be verified in the target portal schema. ReportId requires DeviceName and Timestamp for event identity.

REQUIRED AND ENRICHMENT FIELD MODEL

CATEGORY	FIELD	PURPOSE	REQUIREMENT
Registry event	DeviceRegistryEvents	Registry creation and modification records.	REQUIRED
Key	RegistryKey	Current-user Windows Run key suffix.	REQUIRED
Value name	RegistryValueName	BeaconBrowserHijack.	REQUIRED
Stored path	RegistryValueData	Executable path for triage.	REQUIRED
Host	DeviceId, DeviceName	Asset pivot.	REQUIRED
Event identity	Timestamp, ReportId	Unique record pivot.	REQUIRED
Process	InitiatingProcessFileName, InitiatingProcessUniqueId	Writer identity.	OPTIONAL
Hash	InitiatingProcessSHA1	Writer file pivot.	OPTIONAL

Data quality requirements

- Confirm Windows onboarding and registry sensor health.
- Verify exact key/name/data population with a controlled write.
- Run a negative test using a different value name.
- Review historical collisions without a rarity threshold.

Known failure conditions

- Endpoint is not onboarded.
- Registry telemetry is absent or excluded.
- The actor renames the value or uses only shortcut hijacking.
- Retention expires before hunting.



DETECTION LOGIC

Primary-platform logic with every condition traceable to evidence.

HUNTING QUERY

DETECTION OBJECTIVE

**Find the source-observed
PATCHCORD Run-key value.**

Select registry events only when the key terminates in the current-user Windows Run path and the value name equals BeaconBrowserHijack.

ANALYTICAL LOGIC

Direct filters and investigable output.

Two direct filters and investigable output. No IOC, hash, join, sequence, aggregation, count, rarity, regex or ActionType assumption.

KQL · UNTESTED HUNT

```
// Hunting query - Untested implementation sketch
// Primary platform: Microsoft Defender XDR / Advanced Hunting
// Required telemetry: Microsoft Defender for Endpoint registry events on Windows
// Evidence: Acronis TRU observed PATCHCORD writing its executable path to the
// BeaconBrowserHijack value under the current user's Run key.
// This campaign-specific artifact is a persistence lead, not attribution proof.
DeviceRegistryEvents
| where RegistryKey endswith @"\\Software\\Microsoft\\Windows\\CurrentVersion\\Run"
| where RegistryValueName =~ "BeaconBrowserHijack"
| project
    Timestamp,
    DeviceId,
    DeviceName,
    ReportId,
    ActionType,
    RegistryKey,
    RegistryValueName,
    RegistryValueData,
    PreviousRegistryValueData,
    InitiatingProcessFileName,
    InitiatingProcessFolderPath,
    InitiatingProcessCommandLine,
    InitiatingProcessAccountName,
    InitiatingProcessSHA1,
    InitiatingProcessUniqueId
| order by Timestamp desc
```

STATUS	THRESHOLD	JOIN / SEQUENCE	PRIMARY KEY	DISPOSITION
Untested hunt	None	None	DeviceName + Timestamp + ReportId	Manual triage

What a match means

The endpoint recorded the PATCHCORD-associated startup value and requires immediate file, process and shortcut review.

What a match does not prove

Malware identity, successful execution, shortcut hijacking, C2 activity, compromise scope or actor attribution.



TRIAGE & INVESTIGATION

Evidence collection before escalation.

SOC WORKFLOW

NUMBERED WORKFLOW

- 01

Confirm the raw event.
Verify exact key, value name, data, host and event identity.
- 02

Resolve the stored path.
Collect the referenced file, hash, signer and origin.
- 03

Inspect the writer.
Review process lineage, account, command line and session.
- 04

Inspect shortcuts.
Find browser .lnk backups and changed target paths.
- 05

Scope follow-on.
Review persistence, browser launches, network and nearby endpoints.

Escalation gate

Escalate when the value is unowned, points to a user-writable or unsigned binary, was written by an unknown process, coincides with browser shortcut changes, or is followed by unexplained C2 activity.

Primary benign overlap

A lab replay or internal application could reuse the value name. Tune only with verified software owner, signer/hash, immutable path and change record; never suppress the entire Run key.

INVESTIGATION PIVOTS

Host
Role, owner, exposure and sensor health.

Registry data
Referenced executable path and prior value.

Writer
Process, hash, account, command line and parent.

Filesystem
Installer, implant and browser shortcut backups.

Network
C2 registration and task polling.

Identity
User context and related endpoints.

DISPOSITION CRITERIA

OUTCOME	MINIMUM EVIDENCE	ACTION
BENIGN	Approved, owned and repeatable activity.	Document and tune narrowly.
SUSPICIOUS	Unknown activity without corroborated malicious follow-on.	Deepen endpoint review.
LIKELY MALICIOUS	Unauthorized match plus coherent file, process, network or control-plane evidence.	Escalate incident response.



FALSE POSITIVES & VALIDATION

Operational controls required before promotion.

QUALITY GATE

FALSE-POSITIVE MATRIX

SCENARIO	WHY IT OVERLAPS	TUNING ACTION
Security lab replay	Analysts may reproduce the published artifact.	Require isolated asset and approved test window.
Internal application	A custom tool could choose the same value name.	Require owner, signed binary and change record.
Forensic restoration	IR tooling may restore an observed value.	Require case ID and analyst identity.

Baseline plan

- Search the full available DeviceRegistryEvents retention.
- Inspect every matching raw event and referenced path.
- Record owner, signer, hash and writer for legitimate matches.
- Do not add rarity or count thresholds before measurement.

Validation procedure

- Confirm MDE onboarding and registry telemetry.
- Perform the controlled positive write and cleanup.
- Run the two negative cases.
- Correlate the positive event with file and shortcut ground truth.

TEST CASES

CASE	EXPECTED RESULT
Create the exact value with a benign test executable	Expected match with complete fields.
Create a different value in the same Run key	No match.
Create the same name under a different key	No match.
Remove the exact test value	Review whether removal is visible; no alert assumption.

Hunting

Initial query

Candidate

Current status

Production

Measured precision

Optimized

Stable feedback

Success

The exact controlled write matches with complete event and process context, negative cases do not, and historical collisions are absent or explainable.

Rejection

The registry event is absent/ incomplete or legitimate software frequently uses the exact artifact without a durable discriminator.

Failure

A controlled exact-name Run-key write produces no usable DeviceRegistryEvents record.



ATT&CK, CONCLUSION & REFERENCES

Coverage statement, residual risk and document control.

CLOSURE

MITRE ATT&CK MAPPING

TECHNIQUE	NAME	COVERAGE STATEMENT
T1060	Registry Run Keys / Startup Folder	Directly covers the source-observed current-user Run-key persistence write.

Detection coverage summary

Strong for the exact value on onboarded Windows endpoints; weak for renamed values, shortcut-only persistence and unmanaged systems.

Residual risk

PATCHCORD can alter the value name or rely on rewritten browser shortcuts. File inspection, shortcut parsing, network telemetry and incident response remain necessary.

ANALYST CONCLUSION

A selective campaign hunt, not a durable behavioral alert.

Acronis directly documents the artifact and Microsoft documents the registry schema. This supports a Medium-confidence Hunting query. Artifact mutability, unknown local prevalence and lack of target-environment testing prevent Production classification.

REFERENCES

- Acronis TRU, "PATCHCORD: New malware cluster targets Afghan telecom and South Asian critical infrastructure," 13 August 2026; archive discovered June 2026. <https://www.acronis.com/en/tru/posts/patchcord-new-malware-cluster-targets-afghan-telecom-and-south-asian-critical-infrastructure/>
- Microsoft Learn, "DeviceRegistryEvents table," updated 11 June 2025; accessed 17 August 2026. <https://learn.microsoft.com/en-us/defender-xdr/advanced-hunting-deviceregistryevents-table>
- MITRE ATT&CK, T1060 Registry Run Keys / Startup Folder. <https://attack.mitre.org/techniques/T1060/>

DOCUMENT CONTROL

VERSION	DATE	STATUS	OWNER
v1.0	17 August 2026	Hunting	Detection Engineering

Publication control

Eight A4 pages. URLs are defanged. No customer data, credentials, HTML source or rendering assets are included in the public repository.